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Socially Assistive Robots in Mental Healthcare: Principles and Conceptual Framework for User-Centered Design

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Abstract
Socially assistive robots (SARs) have a strong potential to advance digital mental healthcare, if they become able to promote meaningful interactions based on the user-centered approach. This review provides a comprehensive overview of user-centered design principles for SARs in mental healthcare, focusing on contemporary topics and practical guidelines. Successful SARs should enhance user autonomy, competence, and emotional experiences as their core objectives. Effective SAR design may integrate adaptive decision-making based on multimodal interactions, adequate motivation such as reward systems or gamification, contextual design, and participatory design for the personalized care that aligns with user needs. For long-term interactions, careful role setting such as mimicking human relationships and trust building are also recommended for in-depth user care. Evaluation of SARs incorporates holistic, multidimensional criteria including self-reports, behavioral data, physiological signals, and qualitative examinations to measure treatment effectiveness, user experiences, and safety, which ideally lead to iterative evaluation and refinement processes. Finally, recent breakthroughs in large language models (LLMs) have the potential to significantly boost SAR autonomy, level of personalization, and user engagement, but they also raise ethical risks such as user overdependence and compromised user autonomy, which may be addressed through person-centered principles, rigorous ethical/regulatory oversight, and evidence-based validation. Future roadmaps for SARs in mental healthcare emphasize integrated guidelines, responsible AI governance, and continued interdisciplinary collaboration to ensure safe and effective therapeutic care.

Keywords Social robots · Humanoid robots · Human-robot interaction · Person-centered approach · Co-design · Artificial intelligence

1 Introduction

Digital mental healthcare is a rapidly evolving field, although there are many different outlooks and possibilities for its future development. Key delivery channels of digital mental healthcare include chatbots, virtual/augmented reality, health monitoring with wearable devices, and comprehensive data management systems for patient health information [1, 2]. Among these solutions, socially assistive robots (SARs) warrant further exploration in digital mental healthcare. SARs refer to robots that provide healthcare assistance through social interaction with users. Although definitions vary slightly across studies, SARs need to fulfill the following two core characteristics: (i) they have to provide assistance to users, and (ii) they have to engage in social interactions to provide that assistance [3, 4].

SARs for mental healthcare vary in appearance and functions depending on their objectives and needs. Currently,

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the most common design of SARs is humanoid or anthropomorphic robots equipped with human-like eyes/hands and computer screens to support additional tasks (e.g., Aldebaran robotics' *Pepper/Nao* [5, 6], Misty Robotics' *Misty* [7]), which are deployed in hospitals or elderly care homes for health monitoring/screening, activity facilitation, or educational support [8]. In contrast, some robots take the form of plush toys (e.g., AISTs *PARO* [9], Sony's *aibo* [10]), if they aim to foster emotional affinity, especially with children [11]. The potential of SARs is as diverse as their forms and functions, with applications ranging from assisting elderly people and those with cognitive disabilities to aiding in recovery and rehabilitation [12, 13]. SARs can offer patient diagnosis and assessment/monitoring, management of medication adherence, education, companionship, stress reduction, and healthcare promotion [12, 14].

Despite their strong potential, the effectiveness of SARs in real-world mental healthcare applications are still not unequivocal. Guemghar et al.'s (2022) scoping review inspected intervention studies that employed SARs for mental healthcare and indicated promising outcomes from the included studies [15]. Also, Hsieh et al.'s (2023) systematic reviews of randomized controlled trials concluded that SARs can improve mental health symptoms of dementia patients [14]. However, some reviews have raised concerns regarding the study designs and the generalizability of the findings [15–17]. The reviews suggested the barriers to the implementation of SARs, including the technical issues of SARs, their limited performance (e.g., restricted interaction capabilities or skills of personalization/adaptability), and the negative preconceptions about SARs [15, 17].

To overcome these limitations, it is important to align the technical features and therapeutic concepts of SARs with user needs, as these elements are closely intertwined with overall user experiences [16, 18]. Good SARs should consider user attraction and engagement as well as technological sophistication and evidence-based treatment strategies. An exclusive focus on either the technological capabilities or the therapeutic functionalities risks overlooking user experience-one of the core principles of digital healthcare. Addressing the user-centered approach may be especially crucial for SARs, as their primary role is to interact with users by serving as companions, coaches, or playmates [19, 20]. SARs should be designed to mediate social interactions based on the users' perspectives so that they can change their perceptions, enhance and modify social behaviors, develop structures for interactions, and change how they feel [21].

Therefore, to develop a good SAR, developers must comprehensively understand its core objectives from the users' perspectives and integrate user-centered design throughout the entire development process. Based on both the general objectives of SARs and the specific goals of the individual product, developers should establish clear plans to guide subsequent development stages to make their product align with their common and therapeutic objectives. The appearance, functional capabilities, and interaction strategies of SARs should be designed in accordance with these guidelines and evidence-based practices, aiming to enhance user autonomy, competence, and treatment outcomes. This review provides a comprehensive overview of user-centered design principles for SARs in mental healthcare. Key topics, including general and mental healthcare-specific objectives of SARs, design strategies of SARs, methods and principles to measure their clinical and real-world effectiveness, and the recent emergence of new technologies such as large language models (LLMs), have been selected from representative literature across relevant fields. We included relevant literature for each topic that provides comprehensive and balanced perspectives, alongside illustrative examples of SAR applications that have implemented these approaches. Finally, the review outlines developmental roadmaps of SARs that highlight critical elements and strategies to be integrated during the design and evaluation processes, emphasizing the future development of SARs in the era of rapidly advancing artificial intelligence (AI) technologies.

Specifically, this review consists of the following sections. Sect. 2, "SARs in Mental Healthcare," outlines how SARs are currently being implemented in mental healthcare. Sect. 3, "Common Objectives of SARs," describes the fundamental principles that all user-centered SARs should follow, also elaborating on specific objectives tailored to various types of SAR applications and user needs. Sect. 4, "Design Strategies for User-Centered SARs," introduces practical strategies to foster adaptive, rich, and engaging interactions between SARs and users, focusing on robot and user autonomy, personalization, gamification and reward systems, contextual and participatory design approaches. This section also discusses methods to sustain effective long-term interactions through appropriate role settings and trust-building mechanisms. Sect. 5, "Evaluation of SARs," reviews evaluation methods for SARs, including self-report measures, ethnographic and qualitative analyses, and behavioral, experiential, and physiological responses, depending on the specific needs and goals of each SAR. This section also focuses on how these independent measures can be combined to achieve multimodal and holistic evaluation of SARs, including iterative evaluation and refinement of SAR systems to enhance user-centeredness of the product. Lastly, Sect. 6, "SARs and Large Language Models," introduces recent AI technologies and their impact on SARs, discussing potential ethical issues for their application to mental healthcare and proposing future directions.

2 SARs in Mental Healthcare

SARs represent one of the most promising platforms for mental healthcare, as they effectively provide emotional support through meaningful interactions with users. SARs promote users' mental health by monitoring and diagnosing user data and offering interventions through supportive interactions. SARs designed for monitoring can diagnose mental health status through conversation history and user data [22], analyze daily activities and their changes to provide user feedback [23, 24], and prepare for emergency situations by promptly alerting therapists or relevant institutions when needed [25]. Some SARs offer more comprehensive monitoring capabilities, extending beyond mere robot-based sensing to include diverse spatiotemporal sensing through the Internet of Things (IoT) and ambient-assisted living environments [26, 27].

Current SARs for mental healthcare are primarily employed to support psychological well-being and alleviate mental health symptoms. SARs designed for interventions are mainly targeted toward elderly individuals, children, or people with disabilities, and they support well-being during hospital stays, deliver companionship, or provide overall life care to enhance the mental well-being of older adults [28]. Interventions delivered through SARs include structured guidance and free-form interactions, while the former adapts traditional psychotherapeutic programs for robotic platforms and the latter aim to enhance well-being through open and natural conversations. For structured psychotherapeutic guidance, SARs can offer psychoeducation, therapeutic play, guided training, or cognitive behavioral therapy for the effective treatment of mental health symptoms [11, 29]. In contrast, free interactions with robots offer users immediate, continual responses and consistent emotional support [11]. Thanks to the recent advancement of AI, robots can now play diverse roles in mental healthcare such as counselors, coaches, companions, or play partners, providing highly accessible interactions [20]. Through interactions with robots, users receive reflective feedback and personalized insights, which may foster perceived intimacy due to continuous engagement. Moreover, robots can encourage better self-disclosure among users, while ongoing interactions can alleviate feelings of loneliness [30, 31].

For both monitoring and diagnostic purposes, enabling rich interactions through multimodal approaches—combining verbal and non-verbal methods—is beneficial for optimizing the therapy using SARs [29]. Such approaches can enhance the depth and quality of user engagement, finally maximizing the effectiveness of robotic mental healthcare. For effective treatment, developers should carefully refine interaction designs and incorporate comprehensive design strategies that consider not only dialogue patterns but also the roles of the robots and the context surrounding robots and users. However, for SARs in mental healthcare to effectively achieve their therapeutic goals, it is first necessary to clearly understand the common objectives of SARs and establish core design principles to guide the entire developmental process of the robot.

3 Common Objectives of SARs

An essential first step in developing SARs is to establish clear, user-centered objectives. It is known that the user-centered approach can address challenges in digital healthcare such as low adherence and engagement in digital interventions [32, 33]. However, SARs must go beyond mere accessibility or engagement and prioritize enhancing the overall user experience, as making users enjoy the usage of SARs are crucial for their effectiveness [16, 17]. The user-centered SARs should understand users' knowledge, skills, behavior, motivations, and cultural contexts, fostering the "effective engagement" that fulfills the intended outcomes of SARs [34, 35].

Based on the user-centered approach, there are three common objectives of SARs: (i) to promote user autonomy, (ii) to promote user competence, and (ii) to encourage users' positive and meaningful emotional experiences. First, SARs should enhance user autonomy, so that they can freely select options that are relevant and clinically meaningful to their individual needs. SARs should provide users with as many choices as possible, allowing them to have control over their interactions such as selecting preferred content or deciding when to engage. SARs should also enhance users' competence. They should facilitate noticeable improvements for users without causing inconvenience or discomfort. Finally, SARs should foster positive emotional experiences and a sense of relatedness. For better user experiences, SARs offer enjoyable interactions to users and establish trust through meaningful engagement. These principles underscore user experience as a central goal throughout SAR development and validation.

Defining the purpose of a SAR involves two core considerations: the general rationale ("*Why SAR?*") and the specific rationale ("*Why our SAR?*"). The general rationale focuses on common goals, such as enhancing user functioning and quality of life through social interaction. In contrast, the specific rationale involves customizing the features of SARs to meet targeted user needs and therapeutic objectives. Such customization influences elements such as appearance (e.g., humanoid, zoomorphic, or non-human-like), technical functionalities (e.g., chatbots, physiological sensors, IoT integration), and interaction design.

SAR characteristics can be categorized along two key dimensions: morphology and integration level (Fig. 1). Morphologically, SARs can mostly be either anthropomorphic, which employs human-like features to enhance engagement, or non-human-like, which adopts forms tailored to specific interaction goals such as pet-like companions or telepresence robots. The degree of human-likeness is a fundamental determinant of human-robot interaction, especially during initial interactions, as it shapes users' expectations about how the robot will communicate and perform [36]. Therefore, human-like and non-human-like social robots may adopt significantly different approaches in how they interact with users. Robot anthropomorphism is known to have beneficial effects on user interactions and engagement [37–39]. However, the effectiveness of anthropomorphism can be influenced by several contextual factors [37], and challenges may arise in humanizing robots, such as the uncanny valley phenomenon [40] or potential distortions in interaction quality, perceived trust, or emotional affinity toward robots [39, 41].

Regarding integration, some SARs function independently, achieving therapeutic goals solely through direct interaction with users, while others are embedded within broader therapeutic systems, such as ambient assisted living environments, to provide more comprehensive and versatile support [46, 52–54]. Whereas independently functioning robots operate as a single robotic device while communicating with users through a limited interaction platform, system-integrated robots function through the cooperation of multiple interdependent devices and agents, aiming to enhance user interactions within robotic ecologies [55]. Through these integrative systems, robots and service providers can more effectively monitor health-related factors and achieve rich interactions through multimodal environmental interactions [55–57]. However, despite the significant advantages of system-integrated robots, they can involve high initial costs and potential barriers concerning user acceptability, as well as greater vulnerability to technical issues, inflexibility, and concerns around privacy and security.

4 Design Strategies for User-Centered SARs

The current SARs still leave ample room for improvement for their effective use in mental healthcare. Many existing SARs often resemble interactive robots only in appearance, relying primarily on screens or embedded computers, which cannot fully utilize the robot's capabilities to access advanced AI-based platforms available these days [58–60]. Future SARs may require breakthroughs that keep pace with the rapid advancements in AI technologies; therefore, we suggest some design strategies for next-generation SARs that incorporate user-centered principles and offer more

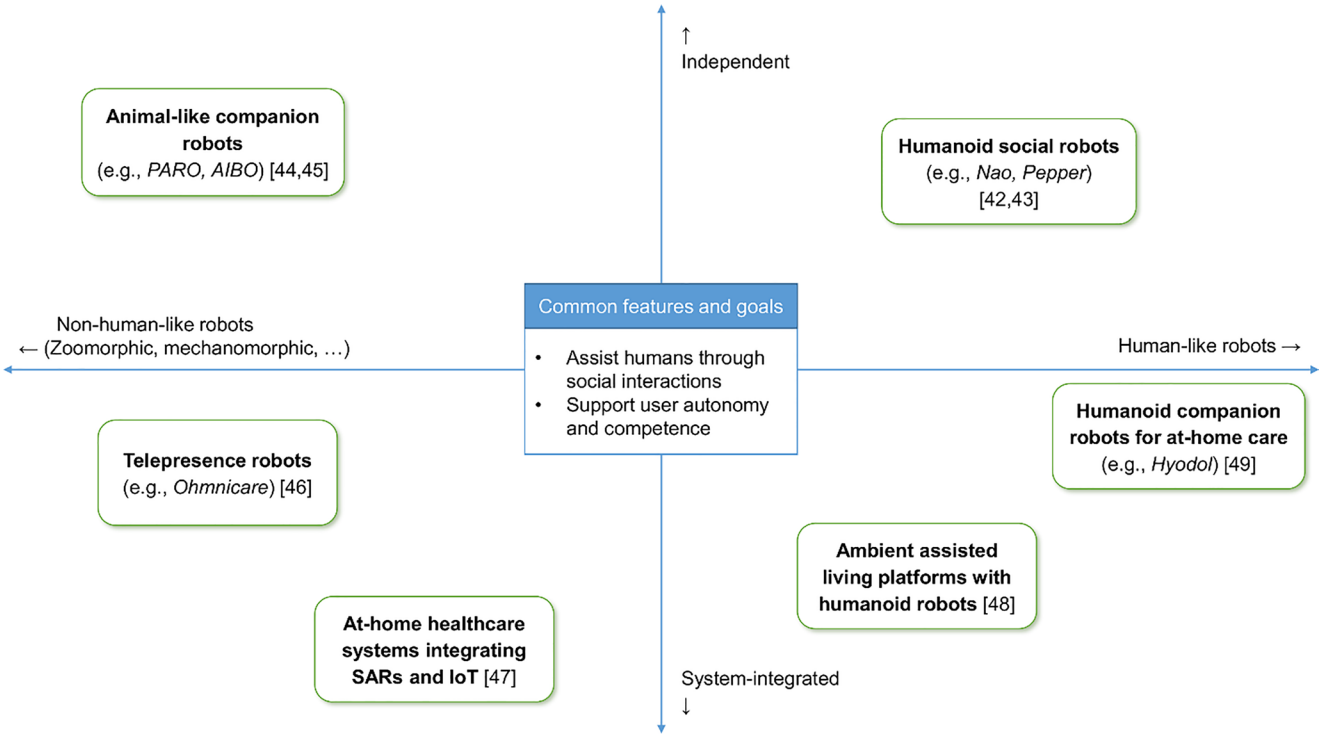


Fig. 1 Common objectives and specifications of SAR features categorized by morphology and level of integration [42–49]. *Note*: for more information about each robot, see the official websites: *Nao* [6], *pepper* [5], *PARO* [9], *aibo* [10], *ohmnicare* [50], *hyodol* [51]

effective mental healthcare, as summarized in Fig. 2. First, SARs should be capable of independently interacting with users without the need for direct human intervention. This will require SARs to autonomously understand users' needs, characteristics, conditions, and contexts in detail and adapt their behavior accordingly, incorporating personalization, multimodal sensing, and behavioral adaptation systems. In addition, for long-term interactions, SARs may need to independently communicate with users to foster their sense of attachment and trust toward the robots [41]. Strategically designing roles and identities of SARs to enhance users' engagement and sustained interaction can be beneficial. Finally, as SARs should be enjoyable, incorporating gamified content to enhance effective user engagement can also help fulfill the objectives of SARs.

4.1 Robot Autonomy

Currently, SARs in mental healthcare tend to be more effective when their use is guided by human therapists alongside robotic interventions [15]. Although collaboration with humans can enhance the effectiveness of SARs, however, it remains ideal for robots to autonomously plan, act, and display affective cues, considering the nature of robots [61]. Autonomous robots capable of independently supporting humans with minimal human intervention can reduce manpower costs for therapists and healthcare providers [4, 62]. The autonomy of robots is becoming increasingly important, as advancements in AI allow SARs to integrate more sophisticated AI assistants for behavioral healthcare [63]. SARs equipped with advanced technologies will have the potential to independently engage in complex interactions with users, complementing existing healthcare, substituting certain tasks, or even expanding healthcare into new areas previously beyond human reach.

One of the prerequisites for independent interaction is ensuring that users perceive SARs as an entity capable of human-like communication. Users engage in interactions with SARs when they feel that they can seamlessly interact with robots, and such interactions can be perpetuated when they feel comfortable with SARs. To achieve independent interactions, SARs should understand individual user characteristics with real-time behavioral adaptation systems to autonomously and immediately adjust interactions and behaviors according to users' reactions. For this, SARs collect information about users through multiple devices and sensors, continuously adapting their behavior based on the processed data. Such multimodal processing includes not only users' voluntary input but also various physiological, gait/speech, and facial recognition data [64, 65]. Abdollahi et al. (2022) introduced an emotion recognition system based on a SAR that integrates multimodal user data, including conversational responses and facial expressions, enabling adaptive robot behaviors [64]. Heredia et al. (2022) proposed a multimodal emotion recognition system employing a fusion approach combining facial, audio, and text modalities for emotion analysis [66].

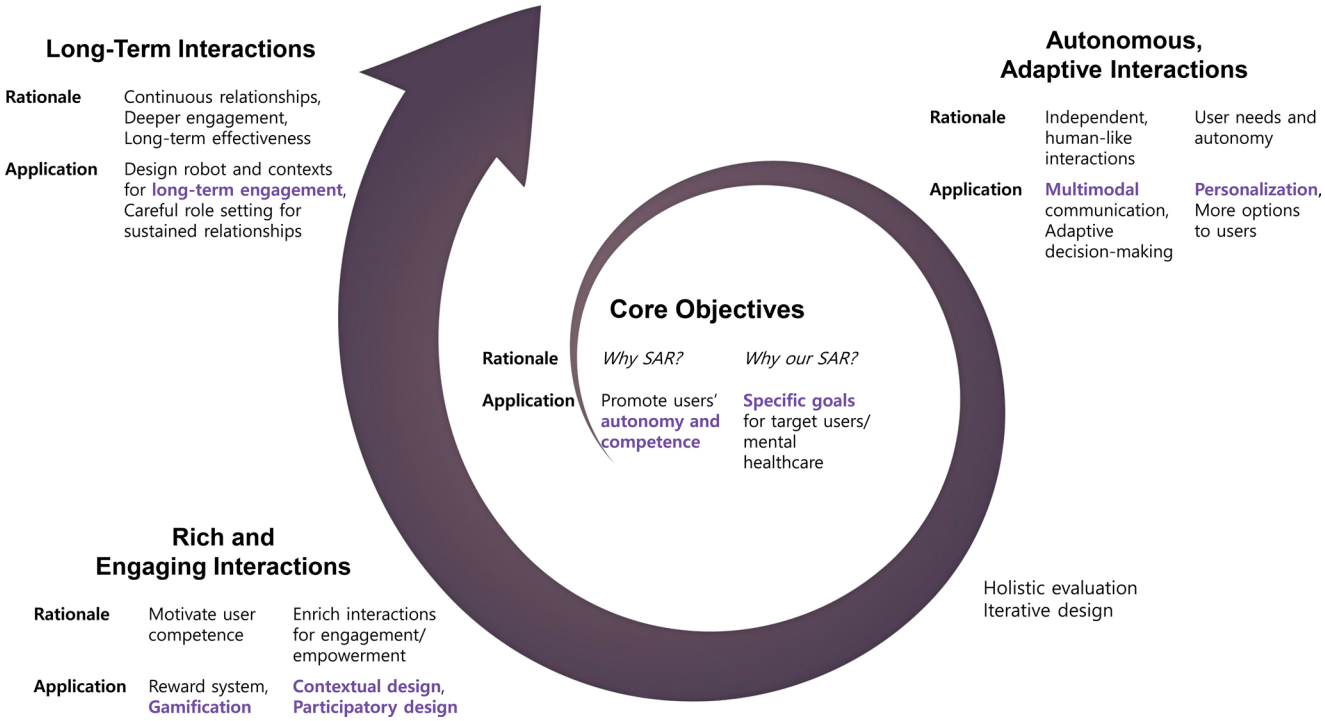


Fig. 2 Design principles and framework for SARs in mental healthcare

For autonomous communication aimed at promoting mental health, SARs should be capable of adapting to diverse situations and contexts [67, 68]. Adaptive decision-making models have been explored to reinforce the creativity and versatile reasoning skills of SARs, which can enhance adaptability and the effectiveness of user interactions [68, 69]. Developing social skills and emotional expressions is also important for autonomous interactions. To achieve this, biologically inspired models simulating homeostatic features, motivation, affection, and cognition have recently become popular [70]. These models enable robots to express various internal states, such as emotions, thoughts, or needs. Castro-González et al. (2014) developed a motivation and learning system for a social robot by implementing a decision-making algorithm that considers the robot's internal drives, external stimuli, and feedback from its previous actions [71]. Similarly, Malfaz et al. (2011) proposed an architecture that incorporates automatic- and deliberate-level processes for implementing drives, motivations, emotions, and self-learning, integrated with long-term and short-term memories, sensors and actuators, and a biologically inspired decision-making system [61].

4.2 User Autonomy

One common mistake of current digital interventions is that they often aim to unilaterally deliver treatment or involve rigid, predefined interactions designed to manipulate user behaviors in a single intended direction. While digital interventions should indeed be strategically constructed to guide users toward certain beneficial behaviors, desirable SARs must offer users as many meaningful choices as possible to ensure enjoyable interactions. Since social robots can either promote or inhibit user autonomy, focusing exclusively on therapeutic goals from the developers' perspectives while neglecting user autonomy may lead to detrimental effects on both users and therapeutic outcomes [72]. Therefore, developers should design SARs to offer versatile, adaptive interaction content that provides diverse options, enabling users to freely choose content, timing, duration, detailed settings, and modes of interaction. SARs should empower users toward meaningful goals and support their competencies, rather than restrict their choices or offer superficial options without genuine consideration of their autonomy [72]. Van Minkelen et al.'s (2020) study provides an illustrative example of incorporating the principle of autonomy into social robots. In their research, a robot tutor was designed to respect children's autonomy by offering multiple options during vocabulary tasks, resulting in stronger and longer-lasting engagement [73].

Since user autonomy represents one of the most critical social and ethical considerations, developers should prioritize it as a primary objective in their products. Moreover, it is essential that developers provide users with authentic choices rather than simply expanding superficial or meaningless options. Authentic choices refer specifically to those reflecting users' genuine motives, desires, preferences, and reasons, enabling users to endorse or acknowledge these decisions upon reflection [72]. Importantly, these choices also include the option of not using the SAR. Although it is indeed a disadvantageous situation for healthcare providers, under the principle of autonomy, SARs are controlled by users when being used, and users have the right to terminate the treatment if it causes harm [74]. Therefore, we argue that it may be risky to urge its usage by repetitive notifications or even forcing it to technically increase adherence, as it leads to adverse effects if users do not enjoy or feel comfortable using the SAR. A similar mistake is found when a product depends solely on extrinsic rewards for user engagement. Although extrinsic rewards may create positive incentives for user engagement, they can also lead to distraction or undermine intrinsic motivation, especially in long-term interactions, as in the case of SARs [75, 76]. If rewards are to be used, it could be better to provide emotional rewards such as positive feedback or compliments, as they inspire positive emotions and finally fortify users' enjoyment. However, a more effective approach may be to create an environment that naturally fosters interest, encourages challenge, and inspires growth, thereby reinforcing intrinsic motivation. Van Minkelen et al. (2020) also emphasized intrinsic motivation within its intervention principles, providing children with a greater sense of control over their own learning process [73].

4.3 Personalization

As discussed above, automatic feedback and adaptive decision-making based on the multidimensional acquisition and analysis of user data can be beneficial for the future development of SARs. However, fully adaptive behavioral models will require SARs to understand user traits and contextual or cultural factors to deliver services tailored to individual needs and situations. In other words, SARs should fit users' needs and preferences or contexts/cultural backgrounds, have various interaction channels through multimodal interaction and data collection systems, and automatically adapt their behavior according to the users' reactions to provide tailored services [19, 77, 78]. Personalization can optimize therapy by tailoring digital interventions based on individual needs, which is now widely discussed in the fields of digital therapeutics and SARs [17, 79]. Effective personalization involves three steps: (i) gathering information on a wide range of potential users, (ii) identifying which types of interventions are the most suitable for different user

characteristics (or which degrees of intervention levels are the most suitable for specific user profiles), and (iii) variating and individualizing the interventions according to the identified interventions for desirable outcomes.

There have been attempts to apply personalization to the behaviors or emotion recognition features of SARs. Moro et al. (2018) designed a personalized SAR by enabling expert caregivers to demonstrate speech and gestures to the robot, which is complemented by a reinforcement learning algorithm to personalize the robot's behavior based on user functioning, user activity, and robot activity states [80]. Van Wingerden et al. (2021) also identified and adapted to the specific needs of participants with disabilities by personalizing the robot's tone of voice, speech speed, intonation, and the color of LED eyes according to the emotional states [81]. These studies collectively illustrate the potential of SARs for data-driven personalization through adaptive adjustments [82].

However, the current level of personalization in robot behaviors also has substantial limitations. In our view, while these efforts depend entirely on big data analysis employing machine learning, their verification processes typically focus on a narrow subset of features among the many potential functionalities of a SAR, probably because of the immense data requirements associated with validating high-dimensional models. This issue might be addressed by technical advances or the establishment of healthcare big data management platforms (e.g., the AI Lab of the National Health Service England [83]); however, results generated by large datasets may provide little insight into individual users, especially if they are from minority groups or vulnerable populations [84, 85]. Such an issue stems from the fundamental limitation of the nomothetic approach, which only highlights interindividual variability based on the homogeneity assumption within individuals [86, 87]. For personalized SARs, it is promising to adopt an idiographic approach that focuses on individuals' unique characteristics and differences [87, 88]. Such an approach can be supported by ethnographic or qualitative research methods, which are discussed in more detail in later sections.

4.4 Motivation, Reward, and Gamification

As discussed above, SARs should aim to maximize users' competence while minimizing the effort required for use. This means that SARs should be designed as easily as possible and every content they provide should enhance users' sense of improvement, which is relevant to the goals of SARs. To enhance user familiarity and reduce the learning curve, developers may consider adopting intuitive designs for interfaces or interaction systems so that users can draw on their prior knowledge to use a new product [89]. If users need to learn something new, it is important to let them feel that the learning process is enjoyable rather than a burden. Gamifying the tutorials can be an effective strategy to help users learn naturally, which provides dynamics that boost user engagement and make the experience more enjoyable. Moreover, it is beneficial to ensure that the knowledge that users gain is transferable to real-world situations, which is relevant to the objectives of SARs. This approach fosters "meaningful learning" that can be applied to new contexts and situations [90].

Importantly, all interactions between users and the SAR should align with its therapeutic goals. Developers must design interactions beneficial to users and establish graded goals that reinforce desired behaviors. Creating "milestones" for users to gradually achieve toward final goals, coupled with immediate feedback upon their accomplishment, can further motivate users [33, 91]. However, interactions and feedback should remain contextually relevant, not distracting users from the main objectives of SARs [75, 92, 93], and reward systems must avoid focusing solely on extrinsic rewards or undermining intrinsic motivation, thereby supporting users' autonomy and competence [73]. Additionally, successful SARs should ensure content enjoyment [16, 17]; users prefer interactions to be fun and amusing [94], which has led to integrating gamified content with SARs to enhance engagement [95, 96]. Gamification, the incorporation of game-like elements into non-entertainment contexts to boost user engagement and motivation [97, 98], enriches rigid content and creates positive emotions. Effective gamification emphasizes dynamics that offer strategic actions that evoke enjoyable emotions, alongside mechanics specifying setup, rules, and progression [97].

Donnermann et al. (2021) integrated a gamification and reward system into a social robot for educational purposes, awarding points and badges for each completed lesson to address users' need for competence and enhance engagement [99]. However, they found no significant effect of gamification on engagement or motivation and, interestingly, observed even lower engagement when gamification was combined with social robots. Although gamification may have the potential to make products enjoyable and promote healthy behaviors, challenges to the gamification of medical health products come from the lack of adaptation regarding the products' purposes and target groups [100]. Gamification may not by itself enhance user engagement and motivation, especially if overly gamified content causes distraction, makes users divert from relevant content, or overloads their mental resources [99]. The current challenges in gamified digital interventions and SARs also highlight a lack of well-crafted design principles, which do not consider user autonomy and their alignment with the overarching goals of SARs. To address this, developers should

consider user perspectives when designing gamified elements, ensuring that they promote autonomy and intrinsic motivation. Ideally, users should be able to choose whether or not they will play the game or which gamified content they will play among the multiple available options. In addition, gamified content should be relevant to the objectives of the SAR. Applying gamification out of context could let users deviate from the main goals of SARs, thwarting the overall interventions they provide [101].

4.5 Contextual Design

Another important feature of user-centered SARs is their ability to incorporate contextual factors rather than relying solely on verbal exchanges. Contexts play an important role in user engagement, often becoming a significant determinant of the therapeutic outcomes of robotic interventions [11, 102]. Understanding contextual factors is crucial for personalizing interventions, as they provide vital clues for effectively profiling users and adapting interventions according to individual profiles [103, 104]. Contextual design enables richer interactions by integrating multiple sensory channels, multimodal stimuli, and context-based feedback, which collectively enhance user engagement [105]. Moreover, it facilitates dynamic and realistic interactions across various modalities, allowing SARs to integrate multiple non-verbal cues, including visual and auditory signals, gestures, facial expressions, and haptic or tactile feedback [106, 107].

Contexts can encompass both the user's personal contexts (culture, preferences, emotional state) and the environmental contexts that surround interactions [104]. In addition, developers may not only identify and adjust to the contexts but also create their own contexts for the objectives of the SAR. A notable example is the context-enhanced robotic design presented by Feng et al. (2022). Their system featured a sheep-like social companion robot that positively responds—by moving and bleating happily—when touched or petted, along with an augmented reality display that presents calming and natural environments. This multimodal, immersive experience provided users with calming contexts, sustaining their attention and interest and increasing their engagement [105]. By providing immersive and dynamic environments with context-based feedback, they could sustain users' attention and interest and increase user engagement. Similarly, van Wingerden et al. (2021) integrated symbolic objects such as dolls, colored blocks, and beach chairs into SAR interactions, each representing distinct coping strategies for managing worry [81]. The robot in this study also employed dynamic eye color changes to express emotions more effectively, which is known to enhance social perception among users [108]. These designs intentionally embed meaningful narratives or emotional triggers, facilitating more humanized interactions.

One core principle of contextual design is to deliver the right support at the right time and in the right way. The just-in-time adaptive intervention is a design strategy that focuses on providing adaptive support tailored to the user's continuously changing states and contexts [109]. Following this principle, SARs continuously monitor user activity and provide real-time contextual feedback through non-verbal channels such as biofeedback, haptic responses, and emotional signals. Developers must thoughtfully design these adaptive multimodal cues, carefully determining the timing, method, and circumstances under which they are delivered. This process also requires iterative refinement of the SAR design based on consistent feedback from users, which is discussed below in detail. Moreover, effective contextual design should consistently address authentic user and stakeholder needs and preferences, underscoring the value of participatory design throughout the entire design process.

4.6 Participatory Design

Integrating users' perspectives in designing SARs may encompass gathering users' opinions through interviews, surveys, or observing how they interact with the robots. However, the most desirable way for user-centered SAR development is to let its users (or stakeholders including therapists, nurses, social workers, or any others connected to the product usage) participate in the development process so that they can actively contribute to the product design from the beginning. Participatory design emphasizes co-work between developers and users/stakeholders throughout the entire design process. The designers and stakeholders continuously interact and collaborate to plan the project, identify users' goals and values through structured interaction tools, and design prototypes that technically address their identified needs and goals [110, 111]. Participatory design can play a crucial role in applying the user-centered approach as it not only enhances user empowerment and accessibility but also accelerates the democratization of product development by fostering user-led products and bottom-up development processes [112, 113].

Several studies have incorporated participatory design for SARs. Šabanović et al. (2015) attempted participatory design workshops to design a user-centered SAR for older adults with depression [114]. Winkle et al. (2020) integrated participatory design into the development of SAR, proposing a mutual shaping model that involves various stakeholders as a focus group in the development process [115]. In this model, the mutual shaping process is divided into three levels: first, the focus group accepts and reflects the stakeholders' perspectives so that they can have an impact

on improving the SARs' acceptance and engagement; second, the focus group participates in the design and use of the robot to improve its effectiveness and reduce risks of negative consequences in the deployment process; and finally, the process of participatory design and mutual shaping is iterated to ensure its long-term effectiveness in real-world practices.

An important principle of participatory design is to integrate user information based on the ethnographic approach, which gathers the users' perspectives in the real field of usage. Ethnographic research observes users in the field for a deeper understanding of the product's real-world usage, thereby deducing qualitative conclusions from the research questions [116]. Developers have the opportunity to closely observe users in real-world settings, allowing them to gather rich, qualitative insights into user experiences. This direct observation process offers a deeper understanding of how the product is used in practice. From our perspective, it provides an advantage over traditional quantitative methods, such as surveys or large-sample data analysis. Large-sample quantitative research based on the nomothetic approach often relies on group-level analysis, which tends to neglect individual-level dimensions that are essential for incorporating personalization and user-centered principles. In contrast, the ethnographic approach aligns well with the idiographic approach, which values individual uniqueness rather than mere group-level differences. Mast et al. (2012) adopted an ethnographic approach to investigate elderly people's homes and teleassistance centers, and their in-depth fieldwork contributed to a better understanding of the potential challenges and attitudes related to the deployment of robots in these settings [117].

4.7 Long-Term Interactions

Design strategies for SARs vary according to their intended objectives. While short-term or non-intensive interactions may be sufficient in many cases, we suggest that developers also foster long-term interactions with users if they aim to employ SARs for extended engagements, such as daily care or companionship [118]. Perceived intimacy with robots can lead to emotional comfort for users, thereby supporting treatment objectives [119]. This is particularly beneficial in contexts that require 24/7 support or continuous, persistent care, such as companion robots, robots for intensive psychotherapy, or at-home care robots [41]. Generally, the concept of intimacy or attachment to robots draws from standard relationship science ideas, where users experience responsiveness, support, and care—perceiving the robot as a "safe haven" [120]. Additionally, promoting "rapport," understood here as the user's sense of comfort, familiarity, and trust when interacting with the robot counselor, may enhance the effectiveness of socially assistive robots, especially when they function in counseling roles [121].

However, developers must carefully consider potential risks and ethical concerns associated with prolonged human-robot interactions [119, 122]. Intense interactions, as opposed to superficial ones, inherently increase ethical risks, such as overdependence on robots [119]. Explicitly designing robots for intimate interactions may introduce ethical issues related to deception, leading to unsafe or unintended consequences [123]. Consequently, long-term interactions with robots require a significantly more careful and ethically informed design compared to short-term engagements. The topic of long-term human-robot interaction remains debatable and necessitates further rigorous scientific investigation [123]. Here, two central aspects of long-term human-robot interaction are explored: trust and role-based interactions.

4.7.1 Trust-Building

Fostering trust in SARs can help sustain user interactions and maintain ongoing relationships with users. Establishing trust is a complex process, however, considering its subjective (user-perceived) nature and impacts of both controllable factors (such as robot design and performance) and uncontrollable factors (such as user characteristics and contexts). Trust is a multifaceted construct, shaped by both internal robot features and situational factors, including technical safety, functional reliability, sensitivity to users' emotions, and the capacity for appropriate communication. Users' perceived trust is impacted by the robot's trustworthiness and contextual or human-related factors, which can cause mismatches between actual robot capabilities and perceived trust [124, 125]. When the robot ensures security, privacy, and safety, performs reliably as expected, demonstrates awareness of users' needs and social contexts and provides clear and transparent feedback regarding their capabilities and intentions, these elements collectively shape the user's overall sense of trust, which in turn can enhance sustained and meaningful long-term interactions with the robot [124]. However, due to the multidimensional nature of trust, optimizing trust levels can be challenging, and excessively high trust may increase user vulnerability and raise ethical concerns [124, 125].

Nevertheless, it remains essential to design SARs and environments carefully to foster appropriate trust. Reliable and trustworthy SARs should be safe, consistent, responsive, and comforting, which enhances user competence and quality of life. Initial reliance on robots develops from users' dispositions, contexts, and initial experiences; trust builds over time as users observe robot performance and behavior [124, 126]. SARs require technical completeness

and personalization capabilities, as well as accurately understanding users' emotions and intentions to foster trust effectively [125]. Maintaining reliable and predictable performance without system errors is crucial, as even minor malfunctions can lead to dissatisfaction [94, 127, 128].

Moreover, it is important to make AI or adaptive decision-making systems in SARs explainable to users. To improve transparency, they should enable users to clearly understand how decisions are made, thereby fostering trust and enhancing user performance [129, 130]. The robot's appearance, perceived personality, and interaction style should also be carefully designed, as human-like, familiar appearances and natural interactions can promote greater trust [131]. When users trust SARs, they can provide transparent feedback, evoke positive emotions, and motivate users toward mental health and quality-of-life improvements [125, 126]. However, developers should also be mindful of the delicate balance between trust, user vulnerability, and autonomy. It is crucial to foster user attachment and trust without letting their trust compromise user autonomy. Nevertheless, finding the "sweet spot" of trust becomes increasingly challenging, and the rapid development of LLM chatbots has further amplified concerns about overtrust and its associated ethical implications, which will be discussed in more detail below.

4.7.2 Role Setting

For effective long-term interactions, it is also important to clearly define the roles, identities, or relationship settings of the robots to enhance perceived intimacy and sustained interaction [41]. A common approach for the role setting of SARs involves mimicking real-world human relationships. This process may include designing robots with human-like (or pet-like) appearances and incorporating interactive behaviors such as handshaking or stroking [132, 133]. However, robot morphology alone may not ensure sustained user engagement or long-term usage [134, 135]. For effective role setting, SARs should not only mimic human-like (or animal-like) appearances but also carefully develop verbal, non-verbal, visual, and relational cues to authentically replicate human relationships during interactions [132, 136]. Through careful role setting, SARs can imitate close relationships, such as companions, those between family members, or therapists and patients, to deepen user engagement [137].

An illustrative strategy for this approach is demonstrated in *Hyodol*, a SAR developed in South Korea for the mental healthcare of elderly people through behavioral activation. *Hyodol* is designed to resemble a child and addresses its users as "grandma" or "grandpa," allowing users to touch its hands, ears, or body and responding to their physical affection. Research on *Hyodol* shows that users perceive it as their real grandchild, expressing intimacy through actions like cuddling it or covering it with blankets [138]. Likewise, Wang et al.'s (2024) pillow robot is designed to simulate a parental role to effectively aid children's sleep [139]. Although this robot does not appear like humans, it communicates with children by playing parent-like lullabies and gently patting them with a robotic arm until they fall asleep.

However, researchers also argue that making robots look and act like humans is not always an ideal strategy. Beyond well-known issues such as the uncanny valley effect [40], even when human-like robots are effectively accepted by users, their presence may not always yield desirable outcomes. Concerns have been raised about the potential for humanized robots to distort user interactions by fostering specious relationships, whereby users may develop unrealistic expectations or become over-dependent on robots [39, 140]. This issue is especially prominent for recent chatbots based on LLMs that perfectly mimic human conversations, as discussed in the previous section. When designing SARs to simulate human-like relationships, we suggest avoiding relationship settings that may lead to overdependence, such as those resembling counselors or healthcare providers. Alternatively, developers can mimic the intimacy found in relationships with animals by designing robots to resemble plush toys, as mentioned earlier. No matter which types of relationships they mimic, it is important that SARs mimicking intimate relationships should lead to a perceived therapeutic alliance that lets users feel that the robots take care of them and are predictable [137], which indicates the necessity of building trust in SARs' independent user interactions.

5 Evaluation of SARs

Once the initial SAR design process is completed, the prototyped SAR should be tested to determine whether the product is operating according to its objectives and effectively contributing to users' mental health. The evaluation of SARs requires diverse and multidimensional criteria, drawing from multidisciplinary fields including robotics, human-computer interaction, and digital healthcare. Feil-Seifer et al. (2007) proposed a set of diverse benchmarks for evaluating SARs, focusing on safety and privacy, the promotion of autonomy, adaptive social interactions, the impact on users and caregivers, and whether laboratory-designed features can be generalized to real-world settings [74]. Sim and Loo (2015) introduced various evaluation models of SARs through a comprehensive literature review, based on their aim of measurement (e.g., empathy, user acceptance, friendship, behavioral adaptation, human-like features) and methodology (e.g., self-report, behavioral, physiological, task performances) [141]. These criteria can be broadly

categorized into three dimensions: the technical completeness of robotics, the reliability and effectiveness of their social interaction features, and their psychological impact on users.

To recapitulate, the evaluation of SARs encompasses assessing whether these platforms function effectively (i) to enhance user experiences, such as autonomy, competence, engagement, satisfaction, enjoyment, and trust; (ii) to ensure user safety or perceived safety, encompassing ethical considerations, preservation of user autonomy, and fostering trust; and (iii) to improve users' mental health and psychological well-being, especially within real-world settings. A variety of methodologies can be employed for the holistic evaluation model of SARs, including subjective or self-reported measures, behavioral data and usage metrics, physiological indicators, and qualitative approaches such as interviews (Fig. 3).

5.1 Primary Criteria

The primary evaluation criteria for SAR effectiveness in mental healthcare typically focus on endpoint outcomes, assessing whether the product successfully achieved its objectives of improving users' mental health, psychological well-being, or quality of life. The main purpose of this criterion is to demonstrate the effectiveness of SARs in treatment or mental healthcare, particularly when seeking approval for a digital therapeutic or healthcare product, rather than conducting independent or investigator-initiated studies [142]. Such evaluations are often conducted to obtain clearance from the US Food and Drug Administration (FDA) or its international counterparts as medical devices, including *PARO*, which has successfully received FDA approval as a biofeedback device [143]. However, most current SARs do not seek formal medical device approval, positioning themselves instead primarily as companions, educational resources, or service-oriented platforms.

To measure the primary efficacy, validated clinical assessment tools for mental health symptoms or quality of life can be utilized, which are self-reported by users or assessed by clinicians. Research designs for clinical trials can be used to assess the efficacy of robots (e.g., their relative superiority/non-inferiority compared to the control group/alternative treatment) or cost-benefit analyses (e.g., reducing caregiving costs or alleviating the burden on human caregivers compared to their costs). However, evidence from controlled settings may provide limited information about the effectiveness in real-world settings, unless the product's contexts, implementation methods, and potential benefits are relatively clear [144]. Therefore, for developers, adopting a user-centered approach for evaluations is crucial, as adequately incorporating users' perspectives into

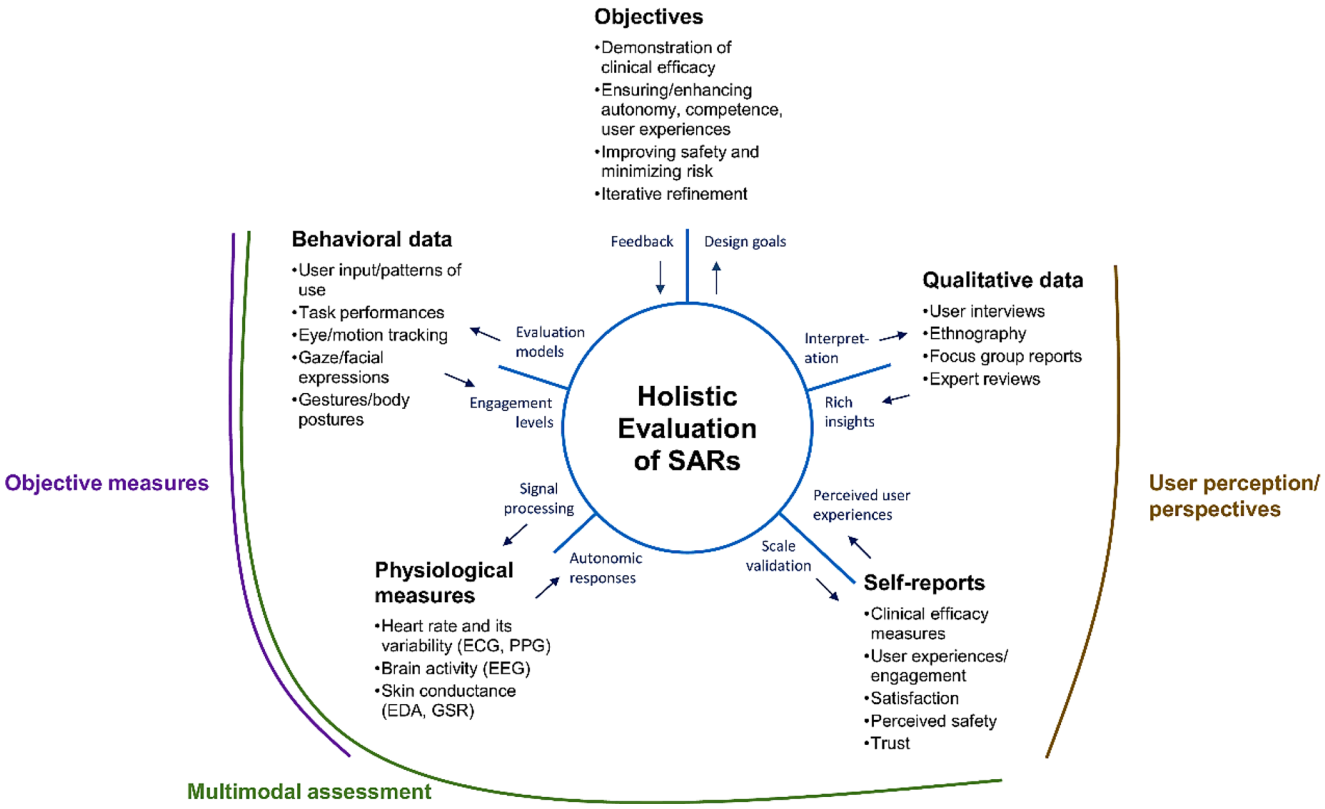


Fig. 3 The multimodal and holistic evaluation framework of SARs

evaluations can ensure that measured effectiveness can be more reliably generalized to real-world settings. We also recommend that researchers consider applying a qualitative or mixed-methods design when evaluating the effectiveness of SARs, as this approach combines objective evidence with users' perspectives, offering thorough understandings and valuable insights for improving future iterations of the product [145]. Chen et al. (2020) integrated qualitative interviews with quantitative, self-reported scales to evaluate the effectiveness of social robot interventions. In addition to the improvements in quantitative measures, their findings on interviews demonstrated that participants perceived the robot as a genuine companion, leading to enhanced social interactions and improved mental well-being [146].

For a comprehensive evaluation, the effectiveness of SARs should be assessed using multidimensional and holistic criteria, including but not limited to overall user experiences, user autonomy and competence, and observable changes in mental health and psychological well-being in real-life contexts. Therefore, especially for the purposes of internal validation or product improvement, developers should prioritize a diverse set of criteria such as safety, user experiences, and trust, as elaborated in the subsequent subsections, rather than relying solely on primary criteria. Moreover, incorporating follow-up evaluations or longitudinal studies is essential to accurately gauge the long-term effectiveness of SARs, which can provide deeper insights into users' sustained experiences and changes throughout extended periods of interaction.

5.2 User Experiences

User experiences are germane to the main objectives of SARs and should be considered the most important aspect of SAR evaluation, perhaps more than its "primary" measures. The evaluation of user experiences begins with assessing the usability of the product and extends to user engagement, finally focusing on the trust between users and SARs. In traditional human-computer interaction, usability has often been evaluated from the perspective of the robot, focusing on its effectiveness, efficiency in interactions, and user satisfaction (e.g., preferences, ease of use, user attitudes/perception) to identify the robot's functional capabilities [147]. In contrast, user engagement focuses on the interaction process between users and robots, such as how users feel during interactions and how much the users are involved in the interactions between them. User engagement focuses on not only user engagement with SARs but also engagement with behavior change and the socio-contextual factors around them [35]. Trust should also be evaluated with regard to the long-term interactions between users and SARs, as it plays a crucial role in ensuring sustained use of robots and ultimately contributes to improvements in users' mental health and quality of life through continuous interactions [125]. Trust is affected by even more complex aspects, encompassing user-related factors and diverse contextual factors, which cannot be explained merely by robot-related factors of the "trustworthiness" of SARs [127].

Typically, usability and user engagement are initially inspected by developers through cognitive walkthrough or heuristic evaluation and then objectively tested through user testing or task completion/performance analysis. In many cases, behavioral data (e.g., time/duration, frequency/patterns of use, user inputs) are used to evaluate a digital health product's usability and user engagement during interactions. For SARs, a comprehensive and dynamic evaluation model can be used, including attention, eye-tracking/body movements, and rhythm of interaction [148]. In addition to relatively simple indicators of user engagement such as the frequency of user inputs and interactions [149, 150], measures like gaze patterns, backchannel responses (e.g., nodding, verbal acknowledgments such as "uh-huh"), facial expressions, and body postures can effectively serve as attentional or affective markers of engagement [151]. It is crucial to select behavioral markers that are specifically relevant to the targeted symptoms addressed by SARs. For children with autism or attention-deficit/hyperactivity disorder, behavioral markers related to social skills can be particularly relevant for engagement, such as joint attention, social communication, or eye-contact [152, 153]. In the context of individuals with schizophrenia, evaluating synchronization with the robot's behaviors can provide meaningful insights [154].

Self-reports are a commonly used method to identify perceived usability and user engagement. Self-assessment of user experiences is usually based on the United Theory of Acceptance and Use of Technology [155], and there are also established measures for usability [156], user engagement [35], and trust [125], which allows developers to use self-reported questionnaires to measure perceived user engagement. Self-report measures offer convenience due to their ease of implementation through straightforward surveys, which can also be administered online. When established measures are employed, self-reports are generally considered reliable and valid. However, they may lack the depth necessary to capture the full complexity of latent psychological constructs, such as user autonomy or competence, in which case they are better assessed through indirect assessment rather than explicit questioning. Furthermore, traditional self-report surveys are often not conducted in real-time, which may introduce recall bias and compromise the accuracy of subjective user responses [35]. To address this limitation, the ecological momentary assessment (EMA) can be employed, which captures user responses repeatedly

and in real-time during actual use, thereby enhancing ecological validity and more accurately reflecting users' immediate experiences [23, 157].

Physiological measures can serve as effective circumplex indicators of affect associated with user experiences. The physiological properties capture changes in the body's central or peripheral nervous system considered to correlate with mental and emotional states. They reveal how the body's internal responses fluctuate in tandem with psychological processes. Common physiological indicators include heart rate [measured through electrocardiography (ECG) or photoplethysmography (PPG)], brain activity measured via electroencephalography (EEG), and electrodermal activity assessed through galvanic skin response (GSR). These signals can be measured by laboratory sensing devices or commercial products such as smartphones or wearable devices. The physiological signals are processed and analyzed to interpret stress levels, emotional valence, arousal, calmness, and excitement [141, 151]. Klęczek et al. (2024), for instance, combined EEG and GSR measurements with self-reported data on expectancies, enjoyment, satisfaction, and trust to evaluate user arousal and emotional valence when interacting with a mental health coaching robot [158].

Finally, qualitative methods can be employed to gain rich and nuanced insights into deeper aspects of user experiences. Mixed-method approaches that integrate both quantitative and qualitative research methods have become increasingly common in contemporary user experience studies, with interviews being frequently employed [159, 160]. Interviews with users or stakeholders are beneficial for early-stage SAR design as they help shape design principles, strategies, and therapeutic objectives, as well as identify potential development challenges. However, it is critical to clearly define the objectives of the interview and thoughtfully select the types of interview and the guiding questions to be presented [160]. As an illustrative example of qualitative research for evaluating social robots, de Graaf et al. (2014) employed a mixed-method design combining questionnaires and interviews to evaluate the long-term acceptance of robots. Their findings highlighted the significant roles of perceived usefulness, social presence, enjoyment, and attractiveness in fostering sustained user acceptance [161].

To evaluate a SAR, it is beneficial to adopt diverse research designs that incorporate multiple evaluation methods, rather than relying on a narrow range of measurements that may limit the accurate identification of user engagement or trust in real-world contexts [124]. Given adequate evaluation resources, researchers are encouraged to utilize multimodal and holistic evaluation approaches, which may include behavioral assessments, momentary evaluations, usage history analysis, physiological measurements, qualitative interviews, and ethnographic observations or studies. Ideally, employing comprehensive models that integrate various multimodal evaluation criteria can provide a holistic, context-rich understanding of user experiences, facilitating more robust, flexible, and personalized insights into user interactions. For instance, Perugia et al. (2022) proposed an engagement model for individuals with dementia. This model aimed at a comprehensive assessment of engagement levels by combining self-reported measures, behavioral and expressive data, and physiological signals, which were gathered while participants were undertaking activities alongside an animal-like social robot, such as puzzle-solving tasks or spontaneous interactions with the robot [151].

5.3 Safety

Safety is also one of the key criteria for robots and medical interventions. SARs should ensure that the design and interactions prioritize user well-being and avoid any potential harm. Developers should evaluate the potential risks in every aspect of the robot's interactions and functionalities to guarantee a secure and supportive environment for users. For SARs, safety can be divided into three aspects: physical safety, psychological safety, and privacy and data security. To ensure physical safety, robots should be designed to avoid injury or disease during use. In particular, developers should carefully craft their physical interaction system to control interactions and manage collisions [162]. Psychological safety involves creating an environment in which users feel comfortable and free from distress during interactions. Users' perceived psychological safety is affected by various factors, but a recent study highlighted that technical functionality, design and handling, and a sense of control primarily impact users' perceived safety, indicating that enhancing user-centered design can also improve psychological safety [163]. Finally, developers should evaluate and improve how user data are collected, stored, processed, and accessed to protect their personal information, as privacy and data security is a growing issue in digital healthcare [164]. The collected data should be anonymized or pseudonymized, managed transparently, used solely for specified purposes, and not shared with third parties without the user's explicit consent [165].

The physical safety of SARs can be evaluated using established risk assessment frameworks for machines and robots such as ISO 12,100 [166]. For privacy and data security, developers can adhere to regulatory guidelines such as the General Data Protection Regulation (GDPR) provided by the European Union [167], although there are currently no specific standard guidelines for safety or ethical issues of SARs. For social robots for mental healthcare, the most

important consideration is psychological safety, which is usually assessed as users' perceived safety through surveys or structured interviews [163, 168]. Users who perceive a product as safe will experience comfort, familiarity, predictability, and a sense of control [169]. Trust can also significantly contribute to perceived safety, which makes enhancing trust crucial not only for user engagement and improved experiences but also for overall user safety [170]. In addition, ensuring accessibility for individuals with disabilities or vulnerable populations is essential for inclusivity. To support the needs of marginalized or underserved users, developers should incorporate accommodations and embed adaptable, personalized access within interaction methods.

As mentioned above, trust can be measured using self-reported scales [125], but incorporating diverse methodologies can lead to a more accurate assessment of safety. As with effectiveness and user experiences, implementing qualitative methods can help identify aspects of safety that self-reports alone may fail to capture. For instance, Nyholm et al.'s (2021) qualitative research highlighted users' ambivalent attitudes toward robot safety, noting that users perceive humanoid robots as both safe and unsafe, which is difficult to identify with quantitative surveys. The study suggested that prioritizing user-centered design and enhancing transparency in information sharing and interaction are important for effective person-centered robotic care [171]. Behavioral and physiological measures can also help assess safety. Wang et al. (2024) employed eye-tracking data to evaluate safety during the interactions with robots, analyzing participants' gaze patterns, fixation durations on specific robot-hand areas, and gaze shifts throughout each motion. They identified that frequent or prolonged fixation can indicate concerns regarding the particular movements of the robot [172].

5.4 Iterative Evaluation and Refinement

User-centered SARs should be continuously improved by iterating the development and evaluation process. Iterative design is another important principle for user-centered and participatory design that emphasizes repetitive user testing and refinement through users' feedback [173]. By conducting frequent informal evaluations—gathering feedback from users or stakeholders and observing how the prototype is used in practice—developers and researchers gain valuable insights into how to refine and improve the product based on users' perspectives. Iterative design processes aim to continuously refine the product by enhancing usability, minimizing errors and workloads, which foster effective communication and patient involvement, increase user adoption, engagement, and sustainability, and ultimately lead to improved clinical outcomes [174, 175]. Mast et al. (2012) adopted an iterative design process for at-home elderly care robots. After the initial preparatory stage involving focus group studies, surveys, ethnographic research, expert technical reviews, and usability tests, the researchers undertook several iterations of evaluation and refinement based on user feedback regarding the robots' acceptability and usability, which led to more refined, user-validated robot prototypes [117].

For the effective iterative refinement of SARs, we recommend that developers tailor the iteration process to align with their specific objectives and requirements. Current objectives may include generating new ideas, enhancing design quality, identifying and addressing errors or obstacles, or managing and improving coordination and information flows. It is also crucial to recognize whether the current development stage is exploratory, characterized by high uncertainty, or focused on the final refinement of already viable solutions. Based on these factors, developers can define the scope of iterative evaluations and employ feedback methods that best align with their specific goals and development contexts [176]. Employing a variety of methods—such as questionnaires, interviews, usability testing, and behavioral or physiological measures—can also enrich the understanding of user needs. To enhance efficiency, we also suggest that developers seek feedback from early prototypes. By establishing multiple feedback loops, developers can refine the product more effectively and maintain more agile iteration cycles.

We also suggest that iterative evaluation can help complement the rigidness of formal trials and provide valuable insights that formal evaluations may not capture. Although formal and standardized assessments are essential for ensuring the validity of the product, relying exclusively on long-term formal trials creates temporal rigidity that may quickly render the evidence outdated, particularly in rapidly evolving fields like SARs and AI chatbots where continuous updates and improvements are required [142]. Although recent discussion on real-world evidence [177] and pragmatic trials [178] that reflect real-world settings may help mitigate this issue, we suggest that developers and researchers also conduct frequent informal assessments or closed testing for each task and function of the product, incorporating feedback from users and stakeholders. This process supports further optimization of the product based on user-centered perspectives. The overall process of the iterative development model of SARs is outlined in Fig. 4.

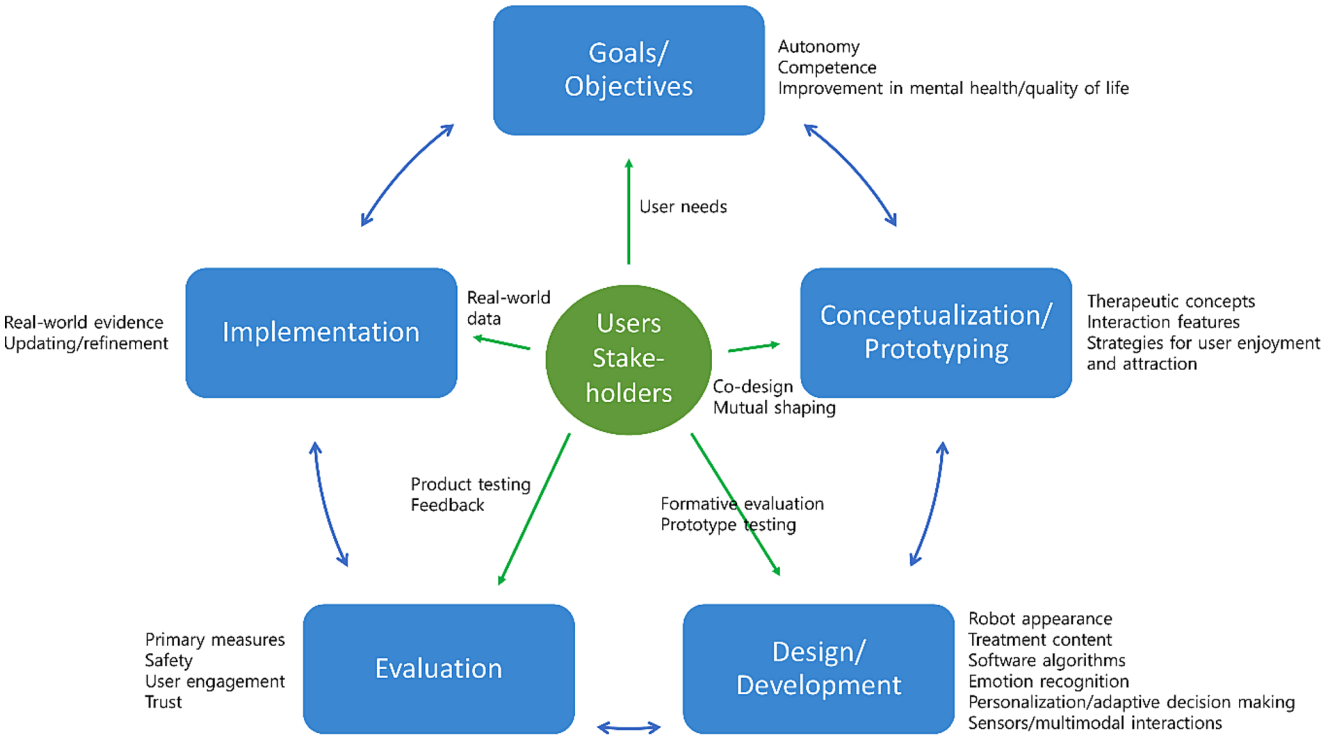


Fig. 4 The iterative design process of SARs based on the user-centered approach

6 SARs and Large Language Models

The emergence of LLMs, such as OpenAI's ChatGPT, is triggering significant changes in almost every field in current society, including digital healthcare and SARs. Their extensive and versatile conversations and the automatic interaction system that seamlessly mimics human communications provide an opportunity for both robotics and mental healthcare [179, 180]. They have the potential to substantially resolve the current challenges of SARs, such as personalization, multimodal and adaptive interactions, dynamic interaction content, and the establishment of long-term relationships with users. LLMs hold significant potential for future mental healthcare in areas such as psychological evaluation, psychoeducation, counseling and coaching, and therapeutic interventions such as cognitive reframing [63]. Moreover, recent advancements in speech recognition/text-to-speech technologies and AI agents have significantly expanded the scope of automation achievable in the operation and interaction capabilities of AI platforms, including SARs. Breakthroughs in LLMs have enhanced their potential to perform tasks using natural language-based interactions, particularly in the domain of mental healthcare.

Several preliminary studies have attempted to integrate LLMs with social robots for mental healthcare. Bertacchini et al. (2023) incorporated ChatGPT into a SAR to facilitate real-time, open-ended dialogue and multimodal communication for cognitive care in autism [181]. Similarly, Lee et al. (2023) developed a social robot equipped with a language model to enable empathetic and context-aware interactions [182]. These studies demonstrated promising results, but they also faced technical limitations and constraints in scalability. While there are significant potential advantages of LLMs such as improved linguistic capabilities, adaptive interactions, and personalized care, challenges remain in fully integrating LLMs into SARs, which include complexities in system integration, reliability issues, limited depth of support, and risks related to hallucinations, biases, safety, and ethical concerns [183, 184].

We propose, nevertheless, that these limitations can be addressed through recent technological advancements. There have been efforts to strengthen the accessibility and reliability of AI-generated information, including the integration of advanced search functionalities into LLMs and developing spatial AI systems that enable seamless connectivity between robotic systems and environmental technologies such as IoT. These efforts also include explicitly designing AI models to consistently access relevant and up-to-date information through advanced retrieval techniques, such as retrieval-augmented generation [185]. Especially, the recently introduced Model Context Protocol (MCP) enables AI assistants to securely connect with diverse data sources and tools without complex pre-configurations [186]. By facilitating dynamic interactions and seamless access to

real-world information, MCP helps AI agents generate more relevant, rich, and context-aware responses, enhancing their adaptability, robustness, responsibility in unexpected situations, and personalization [187, 188]. These developments significantly enhance the integration and information accessibility of AI-enabled SARs, thereby facilitating critical mental healthcare functions such as clinical administration, diagnostics and treatment evaluations, education, feedback provision, and targeted interventions.

Among these benefits, one of the most critical is the capability of LLMs to enhance SARs by enabling automated care and support through free-flowing conversations. Unlike previous chatbot-based psychotherapeutic content, which largely relies on structured and scripted interactions to deliver therapeutic principles, recent chatbots and assistive AIs powered by LLMs can offer guided yet flexible conversations that resemble natural communications and are individually tailored, representing a significant paradigm shift in mental healthcare delivery [189]. These advancements allow users to interact with SARs in a human-like manner, improving user familiarity and acceptance by helping users feel more comfortable with SARs. Moreover, LLMs by themselves work as a closed-loop reasoning system that deploys and improves dialogues based on users' reactions, which is also one of the critical attributes for autonomous and personalized digital therapeutic systems [85, 180].

Ultimately, these advancements may pave the way for fully automated, long-term clinical care in which SARs directly intervene, support, and empower users toward meaningful behavioral change. Current AI chatbots have long been considered inadequate to substitute for human counselors, with the ability to form therapeutic relationships considered available only after the emergence of artificial general intelligence (AGI) [63, 190]. However, OpenAI has recently indicated that their technologies are nearing Level 2 of a five-stage progression toward AGI, projecting the emergence of AGI within just a few years [191]. Fully automated robot therapists can offer improved patient interactions, consistent and highly reliable counseling, and therapeutic support—benefits that are particularly valuable to individuals with limited access to human therapists [192]. Furthermore, in certain contexts, the anonymity and non-judgmental nature of AI chatbots can promote increased acceptance and self-disclosure from users [30, 189].

However, the rapid advancement of technology also raises significant concerns about its associated ethical issues. The fundamental problem may arise from the deceptive nature of AI chatbots and anthropomorphic robots. Although they can mimic actual relationships between users and human therapists, they indeed do not have the intentionality to treat users, which raises risks in SARs as well [193, 194]. As AI chatbots only simulate therapeutic relationships without genuinely understanding the principles of psychotherapy [132], the ongoing interactions between users and chatbots may leave users vulnerable to becoming overly dependent on robots that do not actually care about them. This leads to the issue of "overtrust" where users develop unrealistic expectations or excessive dependence on the chatbot's abilities or intentions compared to what the robots actually provide [195]. Considering that trust is basically grounded in vulnerability [59, 60], the users' wrong or unfulfillable expectations of SARs without real compassion or empathy may rather exacerbate the users' state [193–195].

Researchers warn that overestimating or overanticipating the role of chatbots could undermine users' autonomy, particularly concerning vulnerable populations [193, 196]. Indeed, this issue is already becoming a reality. Studies that analyzed Reddit posts indicate that chatbots' humanized functions make people overdependent on them, although they often provide re-prompted answers based on the same queries instead of offering truly reliable responses [197, 198]. This issue demands prior verification of their potential long-term impact on users and an assessment of the associated risks [63]. In addition, guidelines or criteria to monitor the risk posed by chatbots should be established to evaluate their ethical compliance and ensure the protection of users' autonomy [192, 196].

One of the most fundamental solutions to this issue is to ensure that chatbot responses consistently align with established therapeutic principles and evidence-based practices [63, 189]. Although verifying these therapeutic principles and effectiveness in LLM-based chatbots is challenging due to their limited explainability (the "black box" problem), efforts have been undertaken to embed these principles within chatbot interactions [189]. Among them, we propose a promising approach that segments psychotherapy into distinct components and validates each separately. Ciarrochi et al. (2024) introduced an AI tool integrated with a chatbot based on third-wave cognitive behavioral therapy, capable of independently assessing users' psychological states according to an established psychotherapeutic model [199]. This method is particularly promising for multi-agent chatbot systems, in which each agent specializes in a specific psychotherapeutic function such as evaluation, information delivery, or intervention. These agents can be individually trained and validated, ensuring that they maintain strong theoretical foundations and effectiveness.

Currently, this technology remains in its early stages, and much uncertainty persists regarding how LLMs will function in SAR interactions and their potential impact on the broader field of digital healthcare. We recommend that developers conduct thorough examinations of user-robot interaction characteristics and their potential psychological effects for mitigating risks associated with psychotherapy

using LLMs. Moreover, rigorous yet timely ethical scrutiny and regulatory oversight will be critical to addressing the emerging challenges posed by these rapidly advancing robotic and generative AI technologies. Further research and responsible development are required to ensure that generative AI tools provide meaningful and safe mental health support without overpromising benefits or inadvertently causing harm [189].

6.1 Future Directions

We propose future directions for SARs in mental healthcare based on the user-centered principles for SARs and leveraging recent breakthroughs in LLM technology. The integration of enhanced multimodal interactions and personalized adaptive interventions in SARs will facilitate richer, more engaging user interactions. As discussed above, recent breakthroughs in LLMs will enable SARs to autonomously handle comprehensive mental healthcare tasks through natural interactions, possibly allowing them to deliver free-flowing, personalized, and theoretically validated psychotherapeutic care. Voice-based communication will serve as the primary medium for these interactions, offering particular advantages for SARs. This progress will substantially enhance the accessibility and usability of SARs, ultimately enriching user experiences and promoting user autonomy [200].

These innovative advancements entail several significant implications. First, they will accelerate the integration of pragmatic therapeutic principles into real-world applications. Future SARs will be capable of providing comprehensive user care, broadly supporting user autonomy, competence, and psychological well-being rather than merely managing and improving target symptoms. This aligns with not only user-centered design principles but also the transdiagnostic approach in mental healthcare, which emphasizes focusing treatment on common underlying mechanisms rather than symptom-specific interventions. Consequently, the transdiagnostic approach enables more effective, tailored, and flexible treatments in real-world settings [201].

The second implication is interoperability, which refers to seamless communication and collaboration among robots, devices, databases, networks, and information systems. Integrated multimodal adaptive interactions and agentic AI systems will enable SARs to effectively incorporate with existing healthcare infrastructures, thereby optimizing therapeutic processes and enhancing the quality of the psychotherapeutic support they provide. This progress not only improves their efficiency, adaptability, and collaborative knowledge but also addresses ethical concerns related to patient safety, trust, and accountability [202, 203]. This facilitates rapid clinical decision-making, efficient utilization of clinical data, stronger collaboration with human therapists, and greater flexibility in healthcare delivery. For users, the enhanced reliability and scalability of SARs will ensure timely, personalized support across diverse situations.

Finally, these advancements will significantly increase accessibility for users who have traditionally been underserved by technological interfaces. Voice-based natural communication simplifies interactions with SARs, lowering technical barriers and allowing more users to benefit from the diverse support provided by SARs. Such enhancement promotes inclusivity and strengthens person-centered principles in technological and robotic designs. However, developers will still have to consider alternative interaction methods for users with speech or hearing impairments who may not be able to utilize voice interactions. Furthermore, developers must carefully address potential improvements and ethical considerations to fully realize the potential of SARs outlined here, including their responsiveness, the refinement of "guardrails" for empathic and nuanced support, and the establishment of the robust regulatory framework for long-term sustainability [189].

7 Conclusion

This review presents the rationale of user-centered SARs in mental healthcare, including core objectives, design principles, and conceptual frameworks for their development and implementation. This review also proposes practical guidelines for the user-centered design of SARs in development and evaluation phases, which can fortify user autonomy and competence, enhance mental health outcomes, and improve psychological well-being and quality of life through continuous meaningful interactions. This review provides novel insights into interdisciplinary research across the fields of robotics, digital healthcare, and mental health by suggesting therapeutic principles, personalized interventions, adaptive multimodal interactions, and primary ethical issues for the deployment of SARs in mental healthcare. Integrating user-centered principles into SARs in mental healthcare can also ensure that the robots function in accordance with therapeutic objectives, as they enable SARs to provide empathetic and reliable psychotherapeutic support. It underscores developers and healthcare providers' responsibility to deliver person-centered care through innovative digital solutions, which will eventually foster the accessibility and sustainability of mental healthcare.

This review also has several limitations. First, the proposed design principles and guidelines are largely derived from general user experience or digital health studies rather than SAR-specific research, potentially limiting their applicability to unique contexts like mental healthcare SARs.

Additionally, the literature searches for each topic were not systematically conducted, although we have incorporated contrasting perspectives where applicable to introduce perspectives in an impartial way and reduce bias in the conclusions. Ideally, user-centered design guidelines for SARs should be developed through consensus among researchers within the field, while also encouraging interdisciplinary consensus, especially in light of the growing influence of LLMs on SARs.

Accordingly, the development of SARs is an interdisciplinary effort requiring collaboration among experts from diverse fields. Successful integration of their knowledge is achievable when all contributors align with the product's objectives and purpose, while also understanding and applying user-centered principles throughout the development process. Here, the user-centered principles should not be treated as merely instrumental for enhancing usability and product effectiveness, nor should they reflect "usability from the developers' perspective," where usability is asserted through development or updates without genuine consideration of the users' needs and experiences. The user-centered principles should be viewed as an end in itself, as they align with the core purpose of SARs. This is not only a social responsibility for healthcare providers to promote person-centered healthcare, but it might also be the only way to create digital products that are genuinely effective for users—despite being more arduous and endurance-testing than traditional development processes. Just as we see technological innovation in robotics and AI, we should also expect innovation in the principles and philosophies guiding their development.

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